Appendix A: Preprocessing Steps

CTG Download. We downloaded the CTG results for completed studies with posted results as a JSON export (all available fields) from `https://clinicaltrials.gov/search?aggFilters=results:with,status:com` on August 7th, 2025. During data cleaning, we excluded four records with sample-size errors identified while reviewing candidate type I errors at $\alpha = .01$.

Record Construction. We transformed each ClinicalTrials.gov JSON record into flat, analysis-level rows prior to p-value cleaning and method harmonization. For each study (`nctId`), we iterated over outcome measures with at least one analysis and, for every outcome–analysis pair, emitted one row per denominator count with units "Participants" and with `groupId` in the analysis's declared `groupIds`. We carried forward study metadata (e.g., `briefTitle`, `studyType`), design-information fields (`interventionModel`, `observationalModel`, `timePerspective`, `allocation`), one-hot encoded phases (`EARLY_PHASE1`, `PHASE1`, `PHASE2`, `PHASE3`, `PHASE4`), and MeSH terms/IDs for conditions and interventions from the `derivedSection`. We also flagged whether the study included any references of type `RESULTS` or `DERIVED`.

Analysis Identifier. Each row was keyed by a stable, hash-based identifier,

$$\texttt{analysisId} = \mathrm{MD5}\big(\texttt{nctId} + \mathrm{JSON}(\texttt{outcome};\ \text{sorted keys}) + \mathrm{JSON}(\texttt{analysis};\ \text{sorted keys})\big),$$

which yields reproducible IDs for unique outcome–analysis pairs within a study (independent of JSON key order). This `analysisId` serves as the unit of aggregation in downstream preprocessing (e.g., computing sample sizes and $eJAB$).

Schema. Fields include: `analysisId`, `nctId`, `briefTitle`, `outcomeType`, `outcomeTitle`, `groupDescription`, `units`, `groupId`, `value`, `pValue`, `statisticalMethod`; MeSH fields (`condition_mesh`, `condition_ids`, `intervention_mesh`, `intervention_ids`); `hasResultsOrDerived`; study-design fields; and phase indicators. The field `value` records the number of participants for the given outcome group/arm.

Standardization of Reported p-Values. Reported p-values often included textual qualifiers (e.g., "<0.05"). We standardized them as follows:

1. Flag the presence of a "<" qualifier: `has_less_than` ← `grepl("<", pValue)`.
2. Extract the numeric component via regex:

$$\texttt{pValue} \leftarrow \texttt{str_extract(pValue, "[0-9]*.?[0-9]+")}.$$

3. Drop rows with missing or blank numeric components.
4. Coerce `pValue` to numeric and retain only valid probabilities: $0 <$ `pValue` < 1.
5. Exclude rows with zero sample size, which are incompatible with downstream model mappings.

Harmonization of Statistical Method Names. Free-text `statisticalMethod` labels were mapped to canonical test families using regular-expression rules (`str_detect`). Observations with no match were excluded. The exact mapping rules are:

Listing 1: Regex-based mapping of free-text methods to canonical families

```
statisticalMethod <- case_when(
  # --- T-TESTS ---
  str_detect(originalMethod, regex("\\bt[- ]?test", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("one[- ]?sample", ignore_case = TRUE)) ~ "1-ttest",

  str_detect(originalMethod, regex("\\bt[- ]?test", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("paired", ignore_case = TRUE)) ~ "1-ttest",

  str_detect(originalMethod, regex("\\bt[- ]?test", ignore_case = TRUE)) ~ "2-ttest",

  # --- CONDITIONAL LOGISTIC REGRESSION ---
  str_detect(originalMethod, regex("logistic", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("regression", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("conditional", ignore_case = TRUE)) ~ "clogit",

  # --- LOGISTIC REGRESSION ---
```

```
  str_detect(originalMethod, regex("logistic", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("regression", ignore_case = TRUE)) ~ "glm",

  # --- COX ---
  str_detect(originalMethod, regex("cox", ignore_case = TRUE)) &
    !str_detect(originalMethod, regex("wilcoxon|mantel|signed|rank", ignore_case = TRUE)) ~ "cox",

  # --- LOGRANK / MANTEL-COX ---
  str_detect(originalMethod, regex("log ?rank|mantel[- ]?cox", ignore_case = TRUE)) &
    !str_detect(originalMethod, regex("wilcoxon|signed", ignore_case = TRUE)) ~ "logrank",

  # --- MANN-WHITNEY ---
  str_detect(originalMethod, regex("mann[- ]?whitney", ignore_case = TRUE)) |
    (str_detect(originalMethod, regex("wilcoxon", ignore_case = TRUE)) &
       str_detect(originalMethod, regex("rank", ignore_case = TRUE)) &
       !str_detect(originalMethod, regex("signed", ignore_case = TRUE))) ~ "Utest",

  # --- WILCOXON SIGNED-RANK ---
  str_detect(originalMethod, regex("wilcoxon", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("signed", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("rank", ignore_case = TRUE)) ~ "wilcox",

  # --- KRUSKAL-WALLIS ---
  str_detect(originalMethod, regex("kruskal|wallis", ignore_case = TRUE)) ~ "Htest",

  # --- CHI-SQUARE ---
  str_detect(originalMethod, regex("chi[- ]?square|χ2|chi2|chisq", ignore_case = TRUE)) ~ "chisq",

  # --- REPEATED MEASURES ---
  str_detect(originalMethod, regex("repeated[- ]?measures?", ignore_case = TRUE)) ~ "rANOVA",

  # --- LINEAR REGRESSION ---
  str_detect(originalMethod, regex("linear", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("regression", ignore_case = TRUE)) ~ "lm",

  # --- ANOVA ---
  str_detect(originalMethod, regex("anova|analysis of variance", ignore_case = TRUE)) ~ "ANOVA",

  TRUE ~ NA_character_
)
```

Adjustment for One-Sided p-Values. For explicitly one-sided scalar analyses, we reconstructed the two-sided p-value as

$$p = 2\min\{p_1, 1 - p_1\}.$$

The null-centered one-sided eJAB omits $(n-1)/n$ from the exponent and uses the directional divisor $2(1-p_1)$. NHST instead requires $2p_1 \leq \alpha$. Appendix C.5 gives the method-label rules and full formula.

analysisId Sample Size QC. Certain tests should yield exactly one row per `analysisId` (e.g., one-sample t-test, Wilcoxon signed-rank test). To enforce consistency, we dropped analyses in which such tests appeared with multiple rows under the same `analysisId`.

Aggregation to the Test Level and Parameter Construction. We aggregated records at the `analysisId` (test) level. Fields invariant within a test were taken from the first row—e.g., `nctId`, `pValue`, `statisticalMethod`, `studyType`, phase indicators (`EARLY_PHASE1`, `PHASE1`, `PHASE2`, `PHASE3`, `PHASE4`), `outcomeTitle`, `outcomeType`, `groupDescription`, `conditionIds`, `briefTitle`, and `has_less_than`.
We defined the effective sample size n per test as:

$$n = \begin{cases} \texttt{first(value)}, & \text{one-sample } t\text{-test or Wilcoxon signed-rank;} \\ \max \texttt{value}, & \text{conditional logistic regression;} \\ 0.5 \sum \texttt{value}, & \text{Cox PH or log-rank;} \\ \sum \texttt{value}, & \text{otherwise.} \end{cases}$$

We also set method-specific parameters required by *eJAB*:

$$\texttt{I} = \begin{cases} n_{\text{rows within test}}, & \text{ANOVA, Kruskal–Wallis, repeated measures;} \\ \text{NA}, & \text{otherwise,} \end{cases}$$

$$(\texttt{R}, \texttt{C}) = \begin{cases} (n_{\text{rows within test}},\ 2), & \text{chi-squared test;} \\ (\text{NA},\ \text{NA}), & \text{otherwise.} \end{cases}$$

Mapping to JAB Models and Computation of eJAB. We mapped canonical `statisticalMethod` labels to the `JABO1` model strings required for $eJAB_{01}$. Mapping was performed only for tests included in our analysis (i.e., excluding chi-squared, Cox PH, log-rank, and repeated-measures designs):

Two-sample t-test, paired t-test, one-sample t-test $\rightarrow$ `"t-test"`,
Linear regression $\rightarrow$ `"linear_regression"`,
Logistic regression, conditional logistic regression $\rightarrow$ `"logistic_regression"`,
ANOVA $\rightarrow$ `"anova"`, Kruskal–Wallis $\rightarrow$ `"kruskal_wallis"`,
Mann–Whitney $\rightarrow$ `"mann_whitney"`, Wilcoxon $\rightarrow$ `"wilcoxon"`.

Given $(n, \texttt{pValue}, \texttt{model}, R, C, I)$ for each test, we computed $eJAB_{01}$ via

$$\texttt{JAB} = \texttt{JABO1}(n,\ \texttt{pValue},\ \texttt{model};\ R,\ C,\ I).$$

Exclusions and Final Dataset. We excluded rows with missing `model`, missing `JAB` or `pValue`, and tests with $n = 1$. The manuscript analysis then applied the criteria in Appendix C to obtain the same screened set used throughout: 16,563 eligible primary- and secondary-outcome analyses from 1,790 exact Early Phase I, Phase I, and Phase II trials.

Appendix B: Supplementary Figures

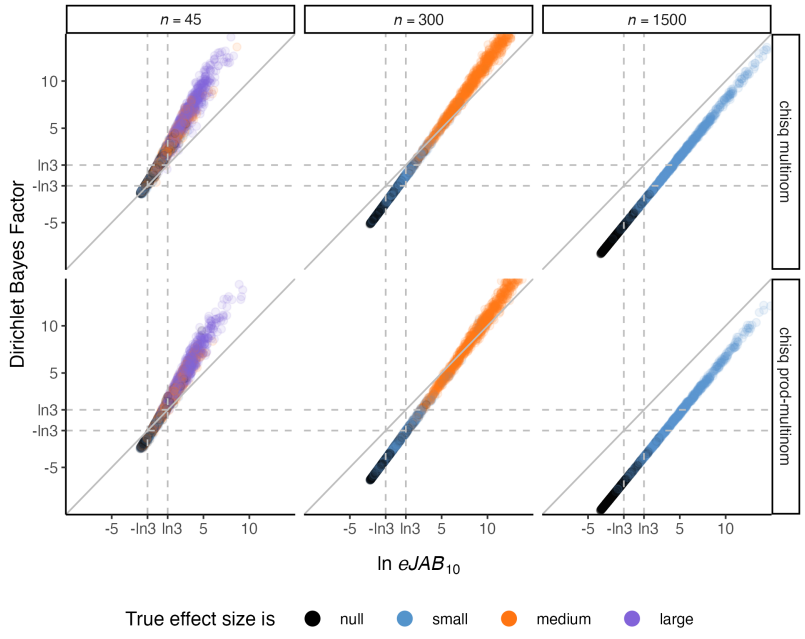


Figure S2: Comparison of the natural logarithmic $eJAB_{10}$ with the `contingencyTableBF` function from the '*BayesFactor*' `R` package at three total counts for chi-squared tests of independence under two sampling designs: multinomial (top) and product-multinomial (bottom). Colors represent the true effect size. The dashed lines indicate $eJAB_{10}$ of 1/3 and 3, representing moderate evidence against and for an effect, respectively.

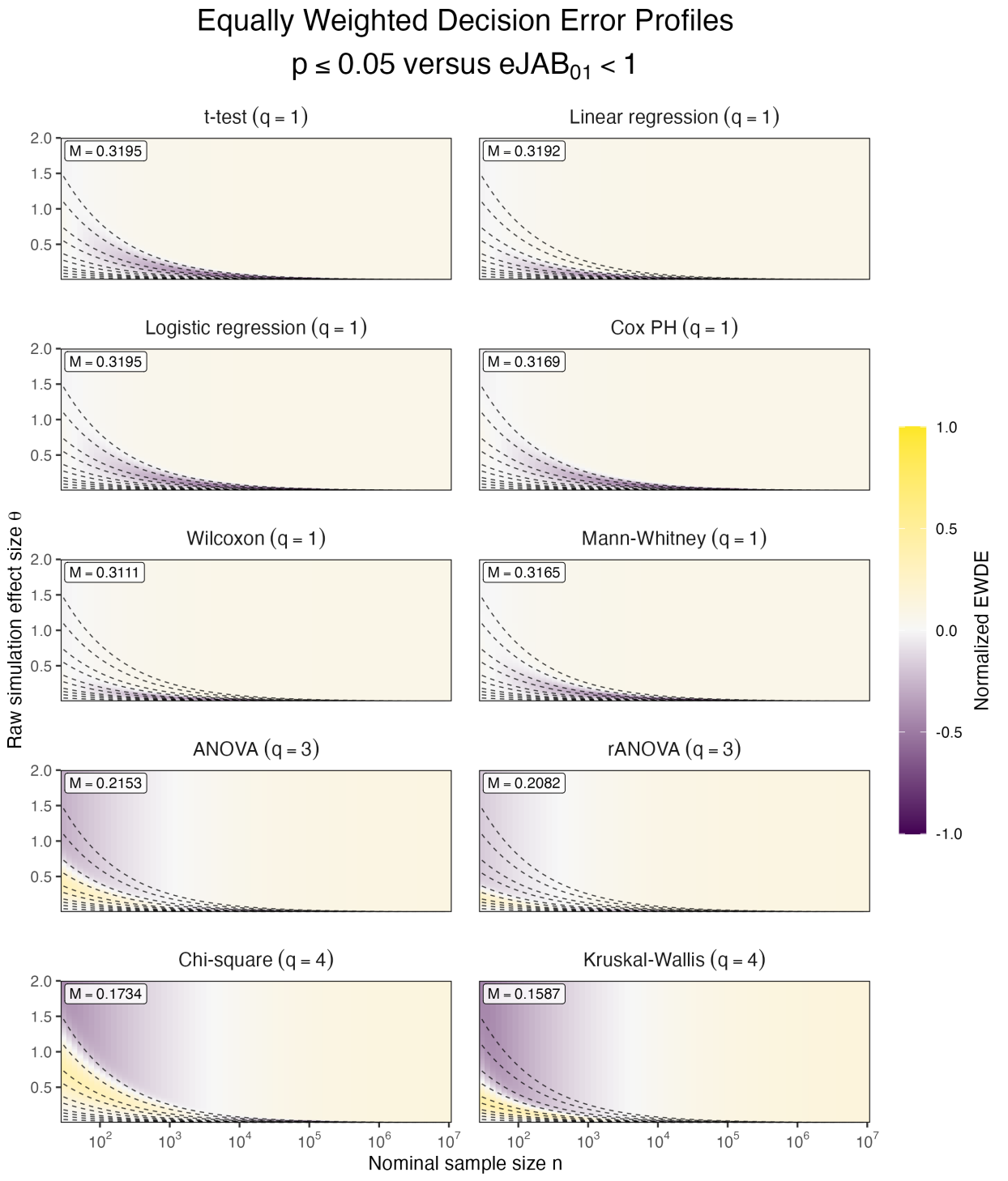


Figure S3: Equally Weighted Decision Error Profiles comparing the NHST significance rule $p \leq .05$ with $eJAB_{01} < 1$ across ten statistical tests. EWDE is the equally weighted average of the type I and type II error probabilities. Colors show $(\text{EWDE}_{\text{NHST}} - \text{EWDE}_{\text{eJAB}})/M$, where each panel reports its maximum absolute EWDE difference M over the displayed grid. Positive values (yellow) favor eJAB; negative values (purple) favor NHST. Dashed curves represent local alternatives of the form $\theta = c/\sqrt{n}$. The corresponding $p \leq .005$ comparison appears in Panel A of manuscript Figure 2.

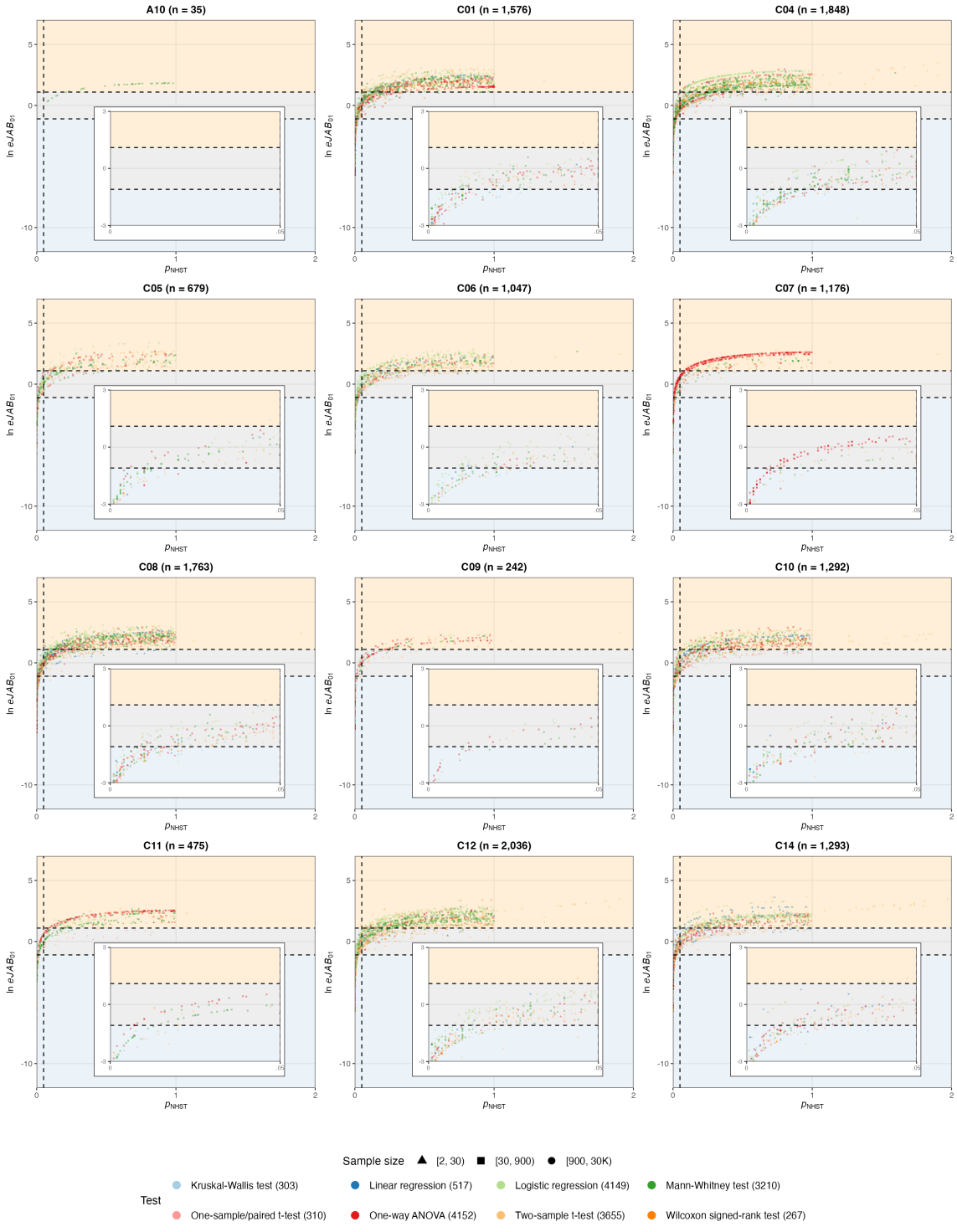


Figure S4: Scatterplots of ln $eJAB_{01}$ versus p_{NHST} by MeSH condition (part 1 of 3), with $p_{\text{NHST}} = p$ for two-sided tests and $2p_1$ for one-sided tests. Insets show $p_{\text{NHST}} \leq .05$. Colors identify tests, shapes sample-size groups, and shading evidence categories.

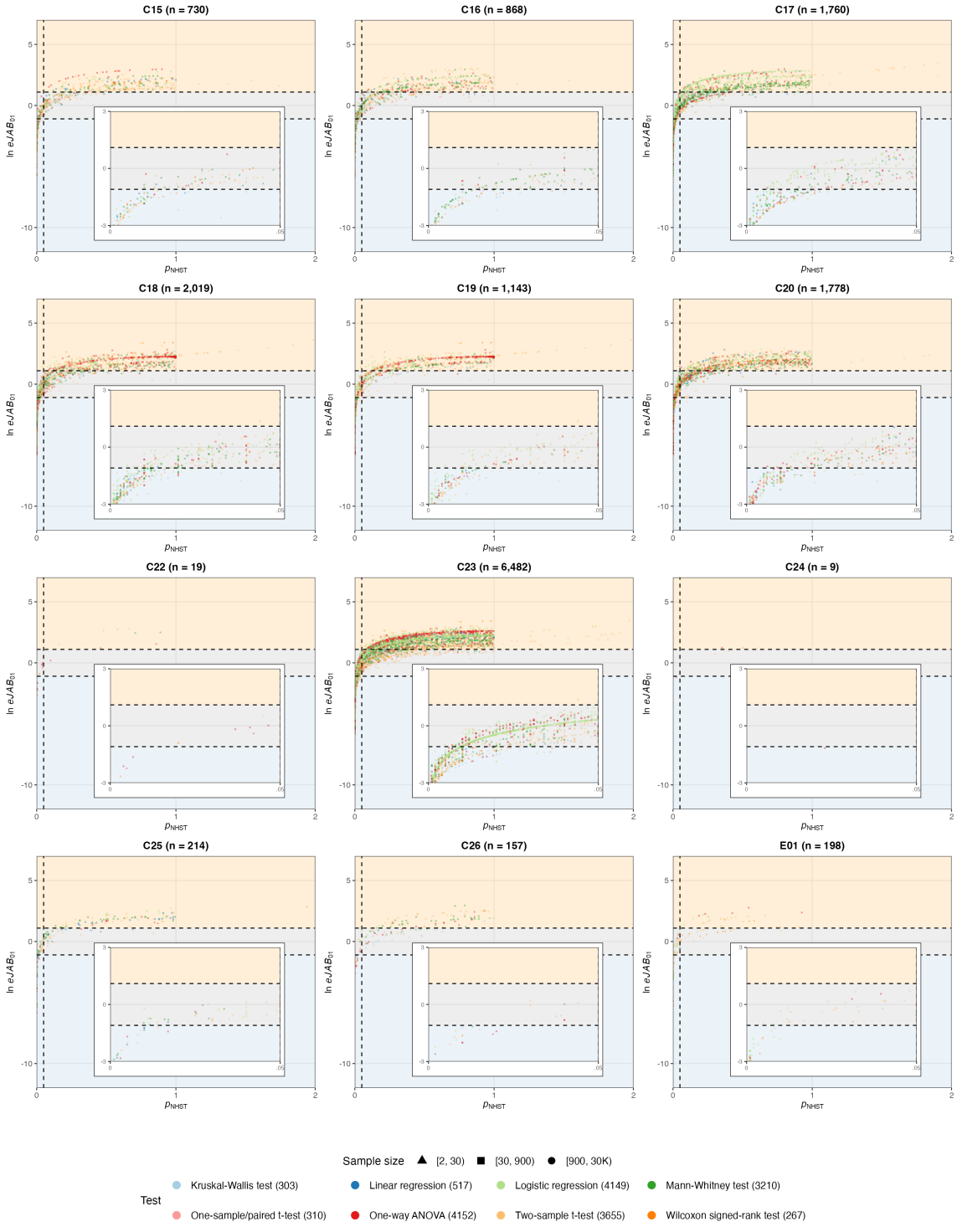


Figure S5: Scatterplots of ln $eJAB_{01}$ versus p_{NHST} by MeSH condition (part 2 of 3), with $p_{\text{NHST}} = p$ for two-sided tests and $2p_1$ for one-sided tests. Insets show $p_{\text{NHST}} \leq .05$. Colors identify tests, shapes sample-size groups, and shading evidence categories.

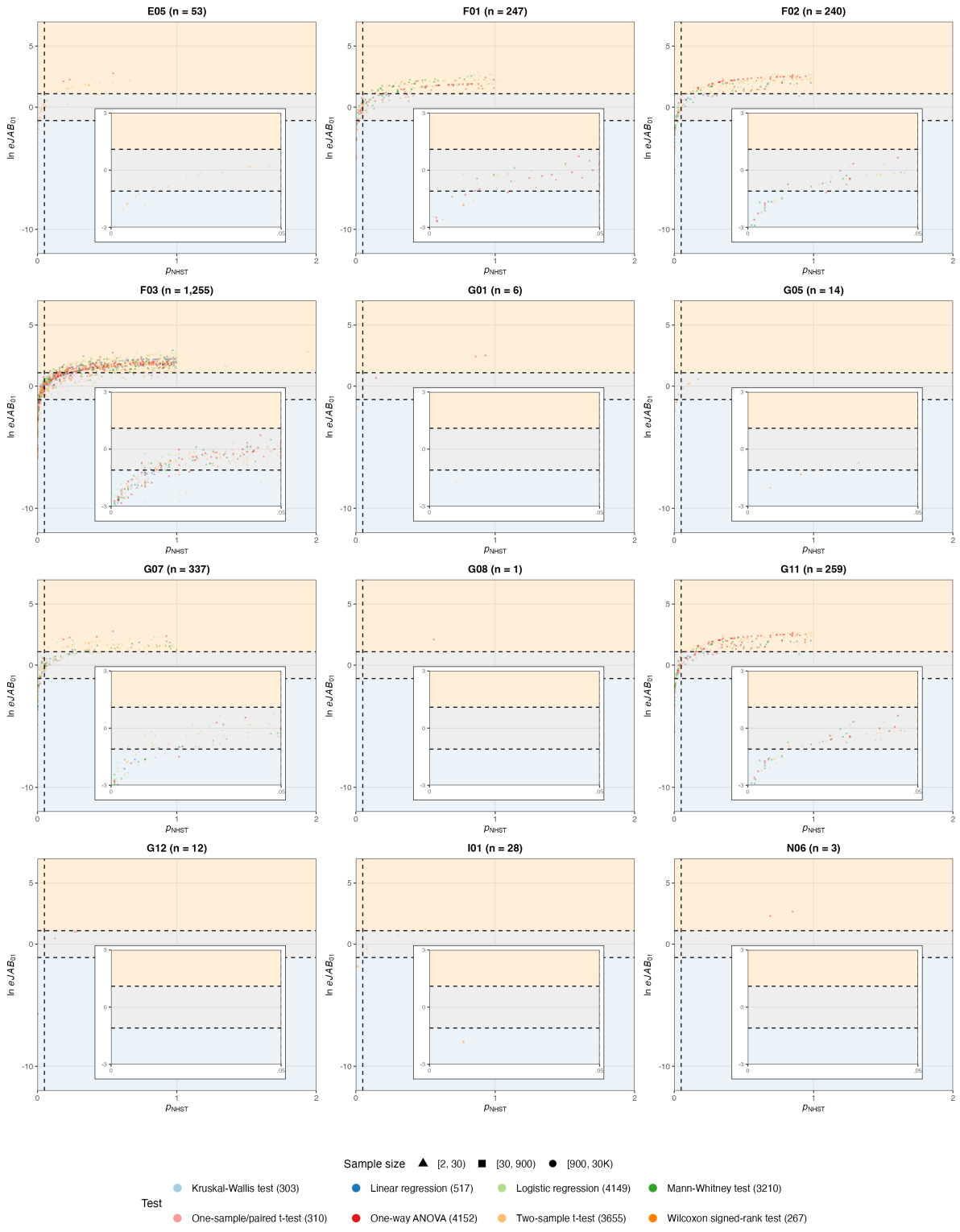


Figure S6: Scatterplots of $\ln eJAB_{01}$ versus p_{NHST} by MeSH condition (part 3 of 3), with $p_{\text{NHST}} = p$ for two-sided tests and $2p_1$ for one-sided tests. Insets show $p_{\text{NHST}} \leq .05$. Colors identify tests, shapes sample-size groups, and shading evidence categories.

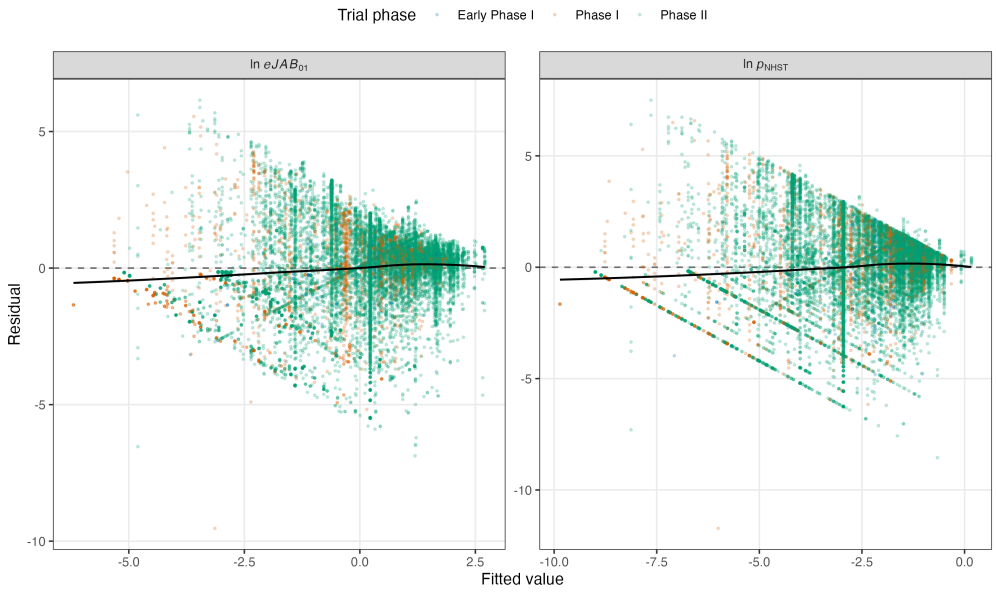


Figure S7: Residuals versus fitted values from random-intercept models for $\ln eJAB_{01}$ (left) and $\ln p_{\mathrm{NHST}}$ (right), where $p_{\mathrm{NHST}} = p$ for two-sided tests and $2p_1$ for one-sided tests, in the screened primary- and secondary-outcome set. The dashed horizontal line marks zero residuals and the solid curve is a LOESS smooth. Nearly parallel slanted bands reflect heaping at common reported p thresholds.

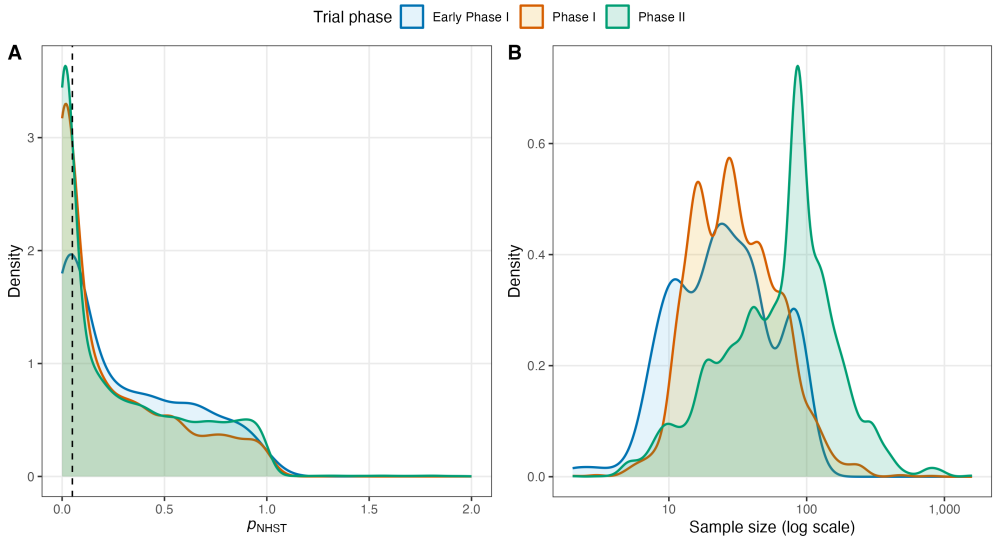


Figure S8: Distributions of p_{NHST} (A; vertical dashed line at .05), equal to p for two-sided tests and $2p_1$ for one-sided tests, and sample size (B) by exact Early Phase I, Phase I, and Phase II classification in the screened primary- and secondary-outcome set.

Appendix C: Detailed ClinicalTrials.gov Inclusion and Exclusion Criteria

C.1 Scope, archived source, and units

The source was the ClinicalTrials.gov (CTG) JSON results export for completed studies with posted results, downloaded on August 7, 2025. The extraction emitted one row for every participant-denominator group attached to an outcome–analysis pair. A deterministic MD5 hash of the CTG identifier, the complete outcome object, and the complete analysis object (with sorted JSON keys) defined `analysisId`. Thus, `analysisId`, rather than an individual denominator row or a CTG study, was the unit of statistical analysis. Rows sharing an `analysisId` were group-denominator components of the same test and were collapsed during sample-size construction. Statistical analyses were nested within CTG studies identified by `nctId`.

The manuscript analysis estimates the frequency of disagreement between NHST significance and eJAB evidence among eligible primary- and secondary-outcome analyses from exact Early Phase I, exact Phase I, and exact Phase II interventional trials. Both one-sided and two-sided/unrestricted tests are included, using $p_1 \leq .025$ and $p \leq .05$, respectively. The three phase classifications are retained as separate groups. These criteria define the screened set used throughout the manuscript and supplement.

C.2 Inclusion criteria

An analysis was eligible only when all of the following criteria were met:

1. The record occurred in the archived CTG results extraction and had a deterministic `analysisId` and `nctId`.
2. CTG classified the study type as `INTERVENTIONAL`.
3. CTG classified the study as *exactly* Early Phase I, *exactly* Phase I, or *exactly* Phase II. Exact Early Phase I required `EARLY_PHASE1`=1 and `PHASE1`=`PHASE2`=`PHASE3`=`PHASE4`=0. Exact Phase I and exact Phase II were defined analogously using their respective indicator.
4. CTG labeled the outcome as `PRIMARY` or `SECONDARY`.
5. The reported p-value contained a numeric value strictly between 0 and 1 and was not right-censored (for example, $p > .05$).
6. The free-text statistical method mapped deterministically to one of the eJAB-computable classes in Table 2.
7. All participant denominators required for the mapped test were finite and positive; the derived sample size satisfied $n > 1$; and the test dimension satisfied $q \geq 1$.
8. The resulting direction-appropriate eJAB was finite and positive.

Statistical significance was *not* an eligibility criterion. Both significant and nonsignificant eligible results were retained for the four-category study-level model.

C.3 Exclusion criteria

The following were excluded, in the sequential order shown in Table 1:

- Four prespecified malformed analysis identifiers previously found to contain denominator errors.
- Observational and expanded-access records, and any record not labeled interventional.
- Phase I/Phase II, Phase II/Phase III, Phase III, Phase IV, not-applicable, or missing phase classifications. Combined phases were not recoded into any of the three early-phase groups.
- Outcomes other than primary or secondary, baseline measures, participant-flow analyses, and adverse-event analyses.

- Missing, blank, nonnumeric, zero, one, or otherwise invalid p-values. Right-censored values were excluded because their numerical bound does not identify the value required for eJAB. Left-censored values were handled as described in Section C.5.
- Methods outside the prespecified mappings, including ANCOVA, mixed-effects or repeated-measures models, Fisher exact tests, chi-squared tests without full table dimensions, Cox or log-rank analyses without the number of observed events, Bayesian analyses, and other free-text methods that could not be mapped without assumptions not supported by the CTG record.
- Analyses with missing, zero, or ambiguous participant denominators; one-sample or signed-rank analyses represented by more than one denominator row; omnibus analyses represented by fewer than two groups; $n \leq 1$; or $q < 1$.
- A nonfinite or nonpositive eJAB value after all preceding steps.

Table 1: Sequential inclusion and exclusion flow. Counts are analysis identifiers except in the last column. Each exclusion count is conditional on all previous steps.

Screening stage	Excluded	Retained	Studies
Unique analyses in the archived extraction	0	279,927	22,454
Remove four malformed analysis identifiers	4	279,923	22,453
Restrict to interventional studies	11,893	268,030	21,500
Restrict to exact Early Phase I, Phase I, or Phase II	175,191	92,839	6,302
Restrict to primary and secondary outcomes	1,806	91,033	6,277
Require an admissible p-value and exclude right-censoring	26,665	64,368	4,461
Require a prespecified eJAB method mapping	47,214	17,154	1,877
Require valid denominators, row structure, n, and q	591	16,563	1,790

C.4 Method mapping, dimension, and sample size

Free-text method labels were normalized using case-insensitive regular expressions. More specific patterns were evaluated before broader patterns (for example, conditional logistic regression before logistic regression, and one-sample or paired t-tests before other t-tests). Table 2 gives the complete set used for the early-phase reanalysis.

Table 2: Eligible statistical method classes and derived quantities. Here G is the number of attached denominator groups.

Mapped class	Sample-size rule	Dimension q
One-sample or paired t-test	One attached denominator; n is that value	1
Two-sample t-test	Sum attached group denominators	1
Linear regression	Sum attached denominators	1
Logistic regression	Sum attached denominators	1
Conditional logistic regression	Maximum attached denominator	1
One-way ANOVA	Sum attached group denominators; require $G \geq 2$	$G - 1$
Wilcoxon signed-rank test	One attached denominator; n is that value	1
Mann–Whitney test	Sum attached group denominators	1
Kruskal–Wallis test	Sum attached group denominators; require $G \geq 2$	$G - 1$

Participant counts containing thousands separators were parsed as numbers before aggregation. A missing or nonpositive attached denominator made the full analysis ineligible; missing values were not silently removed from a sum. The archived extraction's repeated `analysisId` rows encode attached denominator groups and were therefore aggregated, not counted as duplicate hypothesis tests. The hash construction produced one final record per outcome–analysis pair; no additional duplicate analysis identifiers remained after aggregation.

C.5 Reported inequalities, test direction, and Wagenmakers's one-sided Bayes factor

The numeric component of a reported p-value was extracted reproducibly. For a left-censored value such as $p < .05$, the bound .05 was used in the eJAB calculation. This is conservative for evidence against the null because the unknown underlying value is smaller. The reported inequality itself was retained: a report of $p < .05$ met the $p \leq .05$ decision rule even though the stored numerical bound equaled .05. Numerical reports were compared as displayed, so a reported $p = .05$ was also included (and likewise $p_1 = .025$ for a one-sided test). A rounded value at the threshold does not establish that the underlying full-precision value satisfies the rule; these classifications describe the reported values. Values reported with ">" were excluded because a lower bound does not identify the tail probability needed for eJAB. The final eligible set contained 2,714 left-censored values (30 in Early Phase I, 429 in Phase I, and 2,255 in Phase II).

Direction was determined from the reported `statisticalMethod`. The patterns `one-sided`, `1-sided`, `one-tailed`, and `1-tailed` identified a one-sided test. A directional correction was used only for scalar tests ($q = 1$). An explicit two-sided label or the absence of an explicit one-sided method label was classified as two-sided/unrestricted. Mentions of one-sided α found only in sample-size or power prose were not allowed to reclassify the reported test. This rule identified 427 one-sided analyses and 16,136 two-sided/unrestricted analyses.

For a one-sided reported value p_1, the corresponding two-sided value was first reconstructed as

$$p = 2\min\{p_1, 1 - p_1\}.$$

For two-sided/unrestricted analyses, eJAB was computed as

$$eJAB_{01} = \sqrt{n}\exp\left\{-\frac{1}{2}\left(1 - n^{-1/q}\right) Q_{\chi^2_q}(1-p)\right\}.$$

For one-sided scalar analyses, we used the null-centered definition in the manuscript, combining the Savage–Dickey approximation with the directional correction of Wagenmakers (2022, pp. 20–22):

$$eJAB_{0+} = \frac{\sqrt{n}}{2(1-p_1)}\exp\left\{-\frac{1}{2}Q_{\chi^2_1}(1-p)\right\}.$$

This null-centered construction omits the finite-sample factor $(n-1)/n$ from the exponent. The directional correction uses prior mass 1/2 and approximate posterior mass $1 - p_1$ in the predicted direction for moderate samples and a relatively vague local prior. Thus the procedure does not merely substitute a one-sided p-value into a chi-squared formula.

Statistical significance was defined as $p_1 \leq .025$ for one-sided analyses and $p \leq .05$ for two-sided/unrestricted analyses. All clinical comparisons and plots use $p_{\mathrm{NHST}} = 2p_1$ or $p_{\mathrm{NHST}} = p$, respectively. At each sweep value α, a *Bayes/NHST contradiction* requires $p_{\mathrm{NHST}} \leq \alpha$ and the applicable $eJAB_{01}$ or $eJAB_{0+}$ exceeding 1. Thus the one-sided condition is $p_1 \leq \alpha/2$, preserving the specified direction and agreeing with the reported counts at $\alpha = .05$. Values of p_{NHST} above one are retained as scaled comparison values, not interpreted as probabilities. The folded value $2\min\{p_1, 1 - p_1\}$ is used only inside eJAB: using it for NHST would incorrectly admit opposite-tail results. This descriptive term does not assert that the result is a type I error.

C.6 Final analytic set and reproducibility

The final set comprised 16,563 primary- and secondary-outcome analyses from 1,790 trials: 270 analyses from 47 exact Early Phase I trials, 2,098 analyses from 280 exact Phase I trials, and 14,195 analyses from 1,463 exact Phase II trials. Combining one-sided tests with $p_1 \leq .025$ and two-sided/unrestricted tests with $p \leq .05$, 94 Early Phase I, 927 Phase I, and 5,453 Phase II analyses were NHST significant. Six Early Phase I results (6.38% of all significant results in that phase), 28 Phase I results (3.02%), and 459 Phase II results (8.42%) were Bayes/NHST contradictions, with the applicable eJAB > 1. The significant denominators include 4, 27, and 104 one-sided results, respectively; none has one-sided eJAB > 1.

The complete executable pipeline is `analysis/early_phase/early_phase_analysis.R` in the GitHub repository. It writes the sequential flow table, the eligible analysis-level data set, phase and test summaries, posterior draws, and the clustered mixed-model sensitivity result. The four manual exclusions, method regular expressions, direction rules, significance thresholds, denominator rules, and eJAB calculations are encoded directly in that script.

The July 2026 protocol was archived after the August 2025 CTG snapshot had already been downloaded and inspected. Accordingly, this is described as a protocol-aligned retrospective reanalysis, not as a prospective confirmatory study or an independent replication. The phase-specific posterior proportions combine both test types and are conditional on NHST significance. Each study has a Dirichlet$(1, 1, 1, 1)$ prior for its four significance/eJAB categories; posterior category probabilities are weighted by the number of eligible analyses in that study before conditioning on significance. Posterior means are approximately 24.1%, 16.3%, and 18.1% in Early Phase I, Phase I, and Phase II, respectively. They show no monotonic pattern across phases and differ substantially from the raw proportions because the prior adds one pseudocount per category per study. These comparisons are descriptive, and no confirmatory comparison among phases is made.

Appendix D: Proof of Model-Selection Consistency

Lemma 1. *For $x \to 1^-$,*

$$0 < f(x) = -2\ln(1-x) - 2\ln(-\ln(1-x)) < Q_{\chi^2_q}(x).$$

The proof is provided in the GitHub repository (see files within 'Theorem').

PROOF OF THEOREM 2.3. Assume $\mathcal{H}_0$ is true. Let $\delta \in (0,1)$ and $K > 0$ be arbitrary. Note that **R1** and $W_n = Q_{\chi^2_q}(1-p) \xrightarrow{D} \chi^2_q$ by the continuous mapping theorem.
For $n \geqslant N^* = \lceil K^2 \exp\{Q_{\chi^2_q}(1-\delta)\} \rceil$, we have

$$\begin{aligned}
&\mathbb{P}(eJAB_{01} > K) = \mathbb{P}\left(\sqrt{n}\exp\left\{-\frac{1}{2}\frac{n^{1/q}-1}{n^{1/q}}\,W_n\right\} > K\right)\\
>&\mathbb{P}\left(\sqrt{n}\exp\left\{-\frac{1}{2}\,W_n\right\} > K\right)\\
=&\mathbb{P}\left(W_n < -2\ln\frac{K}{\sqrt{n}}\right)\\
\geqslant&\mathbb{P}\left(W_n < -2\ln\frac{K}{\sqrt{N^*}}\right)\\
\geqslant&\mathbb{P}\left(W_n < -2\ln\frac{K}{\sqrt{K^2\exp\{Q_{\chi^2_q}(1-\delta)\}}}\right)\\
=&\mathbb{P}\Big(W_n < Q_{\chi^2_q}(1-\delta)\Big) \to 1-\delta \text{ as } n \to \infty.
\end{aligned}$$

Thus, $eJAB_{01} \xrightarrow{P} \infty$ as $n \to \infty$ under $\mathcal{H}_0$.

Assume $\mathcal{H}_1$ is true. Let $\delta \in (0,1)$ and $\epsilon > 0$ be arbitrary. Lemma 1 gives $0 < f(1-p) < Q_{\chi^2_q}(1-p)$ for $p \to 0$.

$$\begin{aligned}
&\mathbb{P}(eJAB_{01} < \epsilon)\\
=&\mathbb{P}\left(\sqrt{n}\exp\left\{-\frac{1}{2}\frac{n^{1/q}-1}{n^{1/q}}\,Q_{\chi^2_q}(1-p)\right\} < \epsilon\right)\\
>&\mathbb{P}\left(\sqrt{n}\exp\left\{-\frac{1}{2}\frac{n^{1/q}-1}{n^{1/q}} f(1-p)\right\} < \epsilon\right)\\
=&\mathbb{P}\left(\sqrt{n}\ \left(p\cdot(-\ln p)\right)^{\frac{n^{1/q}-1}{n^{1/q}}} < \epsilon\right) > 1-\delta.
\end{aligned}$$

The final inequality is true for n sufficiently large by **R2** and the continuous mapping theorem with $1 - \frac{1}{n^{1/q}} \to 1$.

Thus, $eJAB_{01} \xrightarrow{P} 0$ as $n \to \infty$ under $\mathcal{H}_1$. Together, *eJAB* is model-selection consistent.

□

Appendix E: Realized Values of D_n in Four Replication Projects

Considering **R2** of Theorem 2.3 from a practical standpoint, we examine the realized values of $D_n = -\sqrt{n} \cdot p \ln p$ in four large-scale replication projects. Each project selected a collection of previously published findings and attempted to replicate them under a preregistered protocol, so that each original result was paired with an independently conducted replication of the same effect.

- **EERP**, the Experimental Economics Replication Project (Camerer et al., 2016, *Science*), which attempted to replicate laboratory experiments published in the *American Economic Review* and the *Quarterly Journal of Economics* between 2011 and 2014;
- **RPP**, the Reproducibility Project: Psychology (Open Science Collaboration, 2015, *Science*), which attempted to replicate a collection of findings drawn from three high-impact psychology journals;
- **RPCB**, the Reproducibility Project: Cancer Biology (Errington et al., 2014, *eLife*), which considered experiments reported in high-impact preclinical cancer biology papers; we use the project's final analysis (Errington et al., 2021, *eLife*); and
- **SCORE**, Systematizing Confidence in Open Research and Evidence (Tyner et al., 2025, *Nature*), which assembled and considered replication of a large corpus of published claims from the social and behavioral sciences.

For each result in the four projects, we extracted the original p-value and sample size n, and the p-value and sample size of each of its replication attempts. Each result has at least one and possibly several replication attempts. A result is included in the $\mathcal{H}_1$ subset when its original p-value satisfies $p \leq 0.005$ and every one of its replication attempts also reached $p \leq 0.005$. Given that the effect was found at this level in the original study and in every replication, we make the assumption that $\mathcal{H}_1$ is true for these results. This gives realized values of D_n under $\mathcal{H}_1$: one from each original study and one from each replication attempt. Table 3 reports the counts at each step.

Two kinds of recorded p-value were treated separately. Where a p-value was recorded as 0, we computed $\ln p$ from the reported test statistic and, where the test has them, its degrees of freedom. Of the plotted values, 51 were obtained this way, including the value at the largest sample size; 12 values (six SCORE original results and six SCORE replication attempts) could not be computed and were dropped. Where a p-value was reported as a bound, such as $p < .001$, rather than as a value, we removed it. Such bounds occur only in SCORE. SCORE labels its original p-values; we removed the 37 it labels as less-than or threshold values, and the one with no label. SCORE does not label replication p-values, so we treated a replication p-value recorded as an exact power of ten, such as .001, as censored unless we were able to recompute the p-value directly from the reported test statistic. This removed two replication values where the p-value is left censored and cannot be recalculated from the reported test statistic.

The columns of Table 3 are as follows.

(1) *Released*: the number of results in each project's released data. A result is a study for EERP (18) and RPP (167, of which 100 were replicated), an effect for RPCB (158 effects from 23 papers, given in 188 rows), and a claim for SCORE (331 claims with at least one replication report).
(2) *With p and n*: the results in column 1 for which both the original p-value and the original sample size are recorded.
(3) *Original $p \leq 0.005$*: the results in column 2 whose original p-value satisfies $p \leq 0.005$.
(4) *$\mathcal{H}_1$ subset*: the results in column 3 for which every replication attempt reached $p \leq 0.005$. A result with no recorded replication p-value is not included; this excludes 20 RPP results and one RPCB result. We also excluded RPP study 69. The RPP data record $p = 10^{-6}$ for both its original and its replication, with no test statistic from which either value could be checked, whereas the reported p-values in the same data are $p < .05$ for the original and $p = .05$ for the replication. For SCORE, a replication attempt is one replication report.
(5) *Plotted values*: the values of D_n shown in Figure S9, one for each original result and one for each

replication attempt in the $\mathcal{H}_1$ subset, less the p-values reported as bounds and the zero p-values whose $\ln p$ could not be computed. For RPCB and SCORE the replication column exceeds the $\mathcal{H}_1$ subset because a result can have more than one replication attempt (up to three in RPCB and four in SCORE), each contributing its own value.

Project	Field	Results				(5) Plotted values	
		(1) released	(2) with p, n	(3) $p \leq 0.005$	(4) $\mathcal{H}_1$ subset	original	replication
EERP	Experimental economics	18	18	7	5	5	5
RPP	Psychology	167	143	67	16	16	16
RPCB	Preclinical cancer biology	158	115	61	15	15	21
SCORE	Social and behavioral sciences	331	311	184	102	58	107
	Total	674	587	319	138	94	149

Table 3: Counts at each step of the selection, by project. The columns are described in the notes above the table.

Figure S9 plots the realized values of D_n against the sample size n, which spans roughly seven orders of magnitude. The most important of the realized values is the one at the largest sample size, which is circled in the figure. Based on this, the condition **R2** seems feasible. The data, code and counts for Table 3 and Figure S9 are in the GitHub repository (see 'Supplementary Information/R2Empirical').

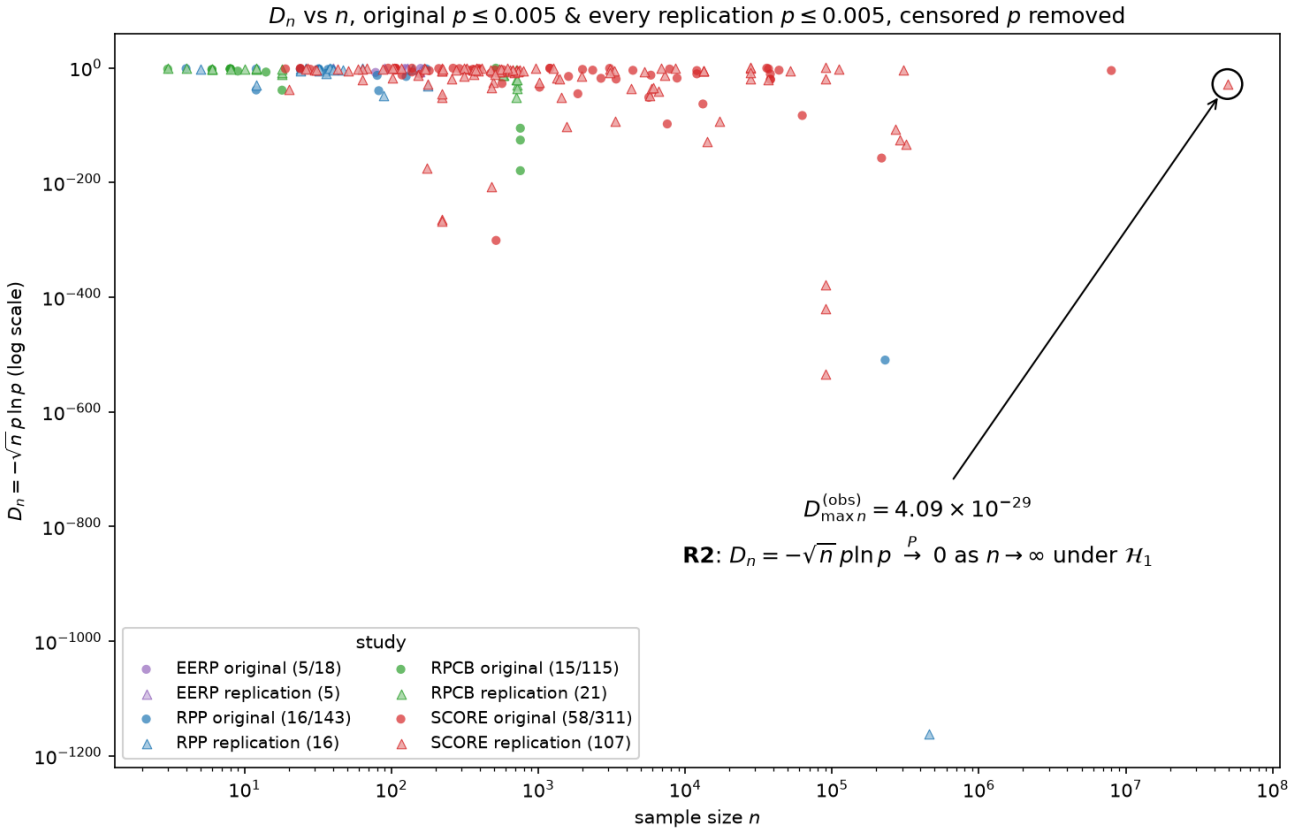


Figure S9: Realized values of $D_n = -\sqrt{n} \cdot p \ln p$, on a logarithmic scale, plotted against sample size n for the $\mathcal{H}_1$ subsets of the four replication projects (original $p \leq 0.005$ and every replication attempt $p \leq 0.005$). Circles denote original results and triangles replication attempts, with symbols colored by project. Legend entries give the number of plotted values; for original results this is shown relative to column 2 of Table 3. p-values reported as bounds are excluded. The circled value, $D^{(\text{obs})}_{\max n} = 4.09 \times 10^{-29}$, is the observed D_n at the largest sample size.